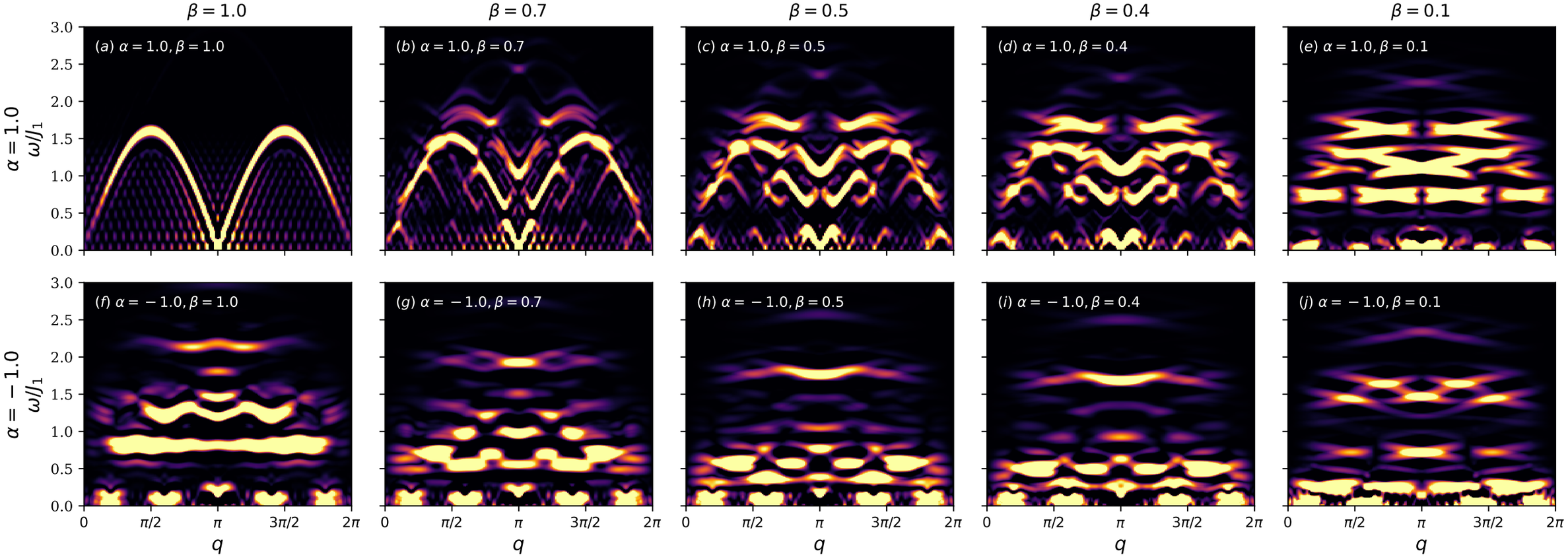

$\beta = 1.0$
$\beta = 0.7$
$\beta = 0.5$
$\beta = 0.4$
$\beta = 0.1$
(a) $\alpha = 1.0, \beta = 1.0$
(b) $\alpha = 1.0, \beta = 0.7$
(c) $\alpha = 1.0, \beta = 0.5$
(d) $\alpha = 1.0, \beta = 0.4$
(e) $\alpha = 1.0, \beta = 0.1$
(f) $\alpha = -1.0, \beta = 1.0$
(g) $\alpha = -1.0, \beta = 0.7$
(h) $\alpha = -1.0, \beta = 0.5$
(i) $\alpha = -1.0, \beta = 0.4$
(j) $\alpha = -1.0, \beta = 0.1$
$\alpha = 1.0$ ω/J_1
$\alpha = -1.0$ ω/J_1
0.0 0.5 1.0 1.5 2.0 2.5 3.0
0 $\pi/2$ π $3\pi/2$ 2π
q
$S^{zz}(q, \omega)$ (mapped)
0.0 0.1 0.2 0.3 0.4 0.5 0.6